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Started on Monday, January 31, 2022, 12:00 AM

State Finished

Completed on Monday, January 31, 2022, 12:01 AM

Time taken 37 secs

Points 50.00/50.00

Grade 100.00 out of 100.00

Question 1

Correct

10.00 points out of 10.00

For the free-falling parachutist with linear drag, assume a first jumper is 70 kg and has a drag coefficient of 12 kg/s. If a second jumper has a drag coefficient of 15 kg/s and a mass of 80 kg, calculate the time taken by him to reach the same velocity the first jumper reached in 9s?

Select one:

- ☐ 16.4150 s
- ☐ 9.0346 s
- ☒ 10.4836 s
- ☐ 12.1124 s



Your answer is correct.

$$v = \frac{mg}{c} \left(1 - e^{-(c/m)t} \right)$$

$$\text{jumper 1 : } v(t) = \frac{9.81(70)}{12} \left(1 - e^{-(12/70)9} \right) = 44.99204$$

$$\text{jumper 2 : } 44.99204 = \frac{9.81(80)}{15} \left(1 - e^{-(15/80)t} \right)$$

$$44.99204 = 52.32 - 52.32e^{-0.1875t}$$

$$0.14006 = e^{-0.1875t}$$

$$\ln 0.14006 = -0.1875t$$

$$t = \frac{\ln 0.14006}{-0.1875} = 10.4836\text{s}$$

The time taken by him to reach the same velocity as the first jumper is 10,4836 s.

The correct answer is: 10.4836 s



Question 2

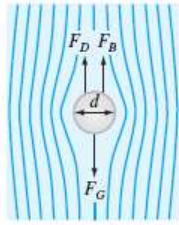
Correct

10.00 points out of 10.00

As depicted in the figure below, a spherical particle settling through a quiescent fluid is subject to three forces: the downward force of gravity (F_G), and the upward forces of buoyancy (F_B) and drag (F_D). Both the gravity and buoyancy forces can be computed with Newton's second law with the latter equal to the weight of the displaced fluid. For laminar flow, the drag force can be computed with Stokes's law,

$$F_D = 3\pi\mu dv$$

where μ = the dynamic viscosity of the fluid (N s/m²), d = the particle diameter (m), and v = the particle's settling velocity (m/s). Note that the mass of the particle can be expressed as the product of the particle's volume and density ρ_s (kg/m³) and the mass of the displaced fluid can be computed as the product of the particle's volume and the fluid's density ρ (kg/m³). The volume of a sphere is $\pi d^3/6$. In addition, laminar flow corresponds to the case where the dimensionless Reynolds number, Re , is less than 1, where $Re = \rho dv/\mu$.



Select one:

☐

$$\frac{dv}{dt} = g(1 - \rho) - \frac{18\mu}{\rho d^2} v$$

☐

$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{18\mu}{\rho_s d} v$$

☒

$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{18\mu}{\rho_s d^2} v$$


☐

$$\frac{dv}{dt} = g - \frac{\rho A}{3m} v|v|$$

Your answer is correct.



A force balance on a sphere can be written as:

$$m \frac{dv}{dt} = F_G - F_B - F_D$$

where,

$$F_G = mg \quad F_B = \rho Vg \quad F_D = 3\pi\mu dv$$

Substituting the individual terms into the force balance yields

$$m \frac{dv}{dt} = mg - \rho Vg - 3\pi\mu dv$$

Divide by m

$$\frac{dv}{dt} = g - \frac{\rho Vg}{m} - \frac{3\pi\mu dv}{m}$$

Note that $m = \rho_s V$, so

$$\frac{dv}{dt} = g - \frac{\rho Vg}{\rho_s V} - \frac{3\pi\mu dv}{\rho_s V}$$

The volume can be represented in terms of more fundamental quantities as $V = \pi d^3/6$. Substituting this relationship into the differential equation gives the final differential equation

$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{18\mu}{\rho_s d^2} v$$

The differential equation for dv/dt as a function of d , ρ , ρ_s , and μ is

$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{18\mu}{\rho_s d^2} v$$

The correct answers are:

$$\frac{dv}{dt} = g(1 - \rho) - \frac{18\mu}{\rho d^2} v$$

,

$$\frac{dv}{dt} = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{18\mu}{\rho_s d^2} v$$

Question 3

Correct

10.00 points out of 10.00

At steady-state, use the result of first section to solve for the particle's terminal velocity.

Select one:

☐ a.

$$v_{\infty} = \frac{g}{18} \frac{\rho - \rho_s}{\mu} d^2$$

☐ b.

$$v_{\infty} = \frac{18}{g} \frac{\rho - \rho_s}{\mu} d^2$$

☒ c.

$$v_{\infty} = \frac{g}{18} \frac{\rho_s - \rho}{\mu} d^2$$



☐ d.

$$v_{\infty} = \frac{18}{g} \frac{\rho_s - \rho}{d^2} \mu$$

Your answer is correct.

At steady-state, the equation is

$$0 = g \left(1 - \frac{\rho}{\rho_s} \right) - \frac{18\mu}{\rho_s d^2} v$$

which can be solved for the terminal velocity

$$v_{\infty} = \frac{g}{18} \frac{\rho_s - \rho}{\mu} d^2$$

This equation is sometimes called Stokes Settling Law.

The particle's terminal velocity is

$$v_{\infty} = \frac{g}{18} \frac{\rho_s - \rho}{\mu} d^2$$

The correct answer is:

$$v_{\infty} = \frac{g}{18} \frac{\rho_s - \rho}{\mu} d^2$$



Question 4

Correct

10.00 points out of 10.00

Use the result of second section to compute the particle's terminal velocity in m/s for a spherical silt particle settling in water: $d = 10 \mu\text{m}$, $\rho = 1 \text{ g/cm}^3$, $\rho_s = 2.65 \text{ g/cm}^3$, and $\mu = 0.014 \text{ g/(cm}\cdot\text{s)}$.

Select one:

- ☐ 5.42321×10^{-5}
- ☒ 6.42321×10^{-5}
- ☐ 6.42321×10^{-4}
- ☐ 5.42321×10^{-4}



Your answer is correct.

Before computing the result, it is important to convert all the parameters into consistent units. For the present problem, the necessary conversions are

$$d = 10 \mu\text{m} \times \frac{m}{10^6 \mu\text{m}} = 10^{-5} \text{ m}$$

$$\rho = 1 \frac{\text{g}}{\text{cm}^3} \times \frac{10^6}{\text{m}^3} \text{cm}^3 \times \frac{\text{kg}}{10^3 \text{g}} = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$\rho_s = 2.65 \frac{\text{g}}{\text{cm}^3} \times \frac{10^6}{\text{m}^3} \text{cm}^3 \times \frac{\text{kg}}{10^3 \text{g}} = 2650 \frac{\text{kg}}{\text{m}^3}$$

$$\mu = 0.014 \times \frac{\text{g}}{\text{cms}} \times \frac{100 \text{cm}}{\text{m}} \times \frac{\text{kg}}{1000 \text{g}} = 0.0014 \frac{\text{kg}}{\text{ms}}$$

The terminal velocity can then computed as

$$v_{\infty} = \frac{9.81}{20} \frac{2650 - 1000}{0.0014} (1 \times 10^{-5})^2 = 6.42321 \times 10^{-5} \text{ m/s}$$

The particle's terminal velocity is $6.42321 \times 10^{-5} \text{ m/s}$.The correct answer is: 6.42321×10^{-5} 

Question 5

Correct

10.00 points out of 10.00

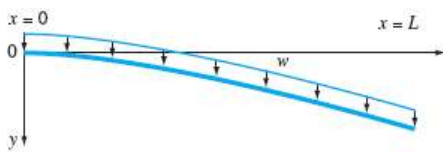
As depicted in the figure below, the downward deflection y (m) of a cantilever beam with a uniform load w (kg/m) can be computed as

$$y = \frac{w}{24EI} (x^4 - 4Lx^3 + 6L^2x^2)$$

where x = distance (m), E = the modulus of elasticity = 2×10^{11} Pa, I = moment of inertia = 3.25×10^{-4} m⁴, w = 10,000 N/m, and L = length = 4 m. This equation can be differentiated to yield the slope of the downward deflection as a function of x :

$$\frac{dy}{dx} = \frac{w}{24EI} (4x^3 - 12Lx^2 + 12L^2x)$$

If $y = 0$ at $x = 0$, use this equation with Euler's method ($\Delta x = 0.125$ m) to calculate the deflection from $x = 0$ to L .



Select one:

- ☒ 0.004819
- ☐ 0.001436
- ☐ 0.004308
- ☐ 0.004204



Your answer is correct.

Substituting the parameters into the differential equation gives

$$\begin{aligned} \frac{dy}{dx} &= \frac{10000}{24 (2 \times 10^{11}) 0.000325} (4x^3 - 12(4)x^2 + 12(4)^2x) \\ &= 2.5641 \times 10^{-5} (x^3 - 12x^2 + 48x) \end{aligned}$$

The first step of Euler's method is

$$\frac{dy}{dx} = 2.5641 \times 10^{-5} (0^3 - 12(0)^2 + 48(0)) = 0$$

$$y(0.125) = 0 + 0(0.125) = 0$$

The second step is

$$\frac{dy}{dx} = 2.5641 \times 10^{-5} ((0.125)^3 - 12(0.125)^2 + 48(0.125)) = 0.000149$$

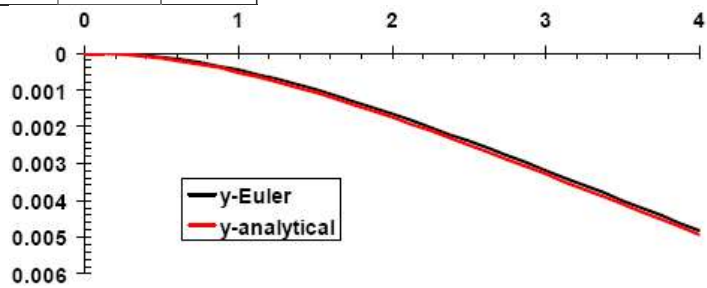
$$y(0.25) = 0 + 0.000149(0.125) = 1.86361 \times 10^{-5}$$

The remainder of the calculations along with the analytical solution are summarized in the following table and plot. Note that the results of the numerical and analytical solutions are close.

x	y Euler	dy/dx
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0	0	0
0.125	0	0.000149
0.25	1.86×10^{-5}	0.000289
0.375	5.47×10^{-5}	0.00042
0.5	0.000107	0.000542
0.625	0.000175	0.000655
0.75	0.000257	0.000761
0.875	0.000352	0.000859
1	0.000459	0.000949
1.125	0.000578	0.001032
1.25	0.000707	0.001108
1.375	0.000845	0.001177
1.5	0.000992	0.00124
1.625	0.001147	0.001298
1.75	0.00131	0.001349
1.875	0.001478	0.001395
2	0.001653	0.001436
2.125	0.001832	0.001472
2.25	0.002016	0.001504
2.375	0.002204	0.001531
2.5	0.002395	0.001554
2.625	0.00259	0.001574
2.75	0.002787	0.001591
2.875	0.002985	0.001605
3	0.003186	0.001615
3.125	0.003388	0.001624
3.25	0.003591	0.00163
3.375	0.003795	0.001635
3.5	0.003999	0.001638
3.625	0.004204	0.00164
3.75	0.004409	0.001641
3.875	0.004614	0.001641
4	0.004819	0.001641



The deflection from $x = 0$ to L is 0.004819.

The correct answer is: 0.004819

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Quiz 1 ▶

